

2.6 Plotting, Ordering, and Comparing Real Numbers

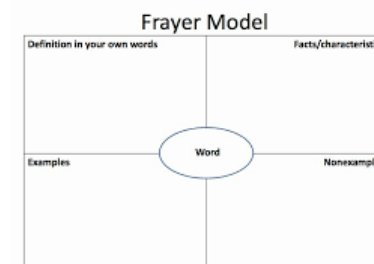
Lesson Introduction

 Benchmark	MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers, represented in various forms.		 Learning Targets	- I can compare rational and irrational values. - I can order rational and irrational values. - I can plot rational and irrational numbers on a number line.
 Benchmark Coherence	Building On MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms, including fractions, mixed numbers, repeating decimals, and percentages, to solve mathematical and real-world problems.			Working Toward 912.NSO1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.
 Essential Questions	- How do you compare rational and irrational values? - How do you order rational and irrational values? - How do you plot rational and irrational numbers on a number line?			
 Lesson Narrative	Students rotate through three stations where they apply what they have learned throughout the unit to order, compare, and plot rational and irrational values. The lesson is designed for students to be able to complete each station without needing guidance from the teacher. The 2.6.1 Warm-Up can be used to revisit common misconceptions before students complete stations, if needed. At the end of the lesson, students reflect on the unit learning targets.			
 Component Summary	2.6.1 Warm-Up (5-10 min.)	Identify irrational numbers, then compare and plot the values (<i>SE p. 91</i>)	2.6.3 Activity (10-15 min.)	Identify rational and irrational numbers and plot them on a number line (<i>SE p. 93</i>)
	2.6.2 Activity (10-15 min.)	Compare and order rational and irrational numbers (<i>SE p. 92</i>)	2.6.4 Activity (10-15 min.)	Identify and correct errors comparing and plotting real numbers (<i>SE p. 94</i>)
	2.6.5 Wrap-Up (~5 min.)	Reflect on understanding of unit learning targets (<i>SE p. 95</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Templates for Frayer Model	
		Additional Preparation		
		2.6.3 Activity has a number line that may need to be enlarged for students who struggle to plot the points in close range.		

 Teacher Tips	Common Misconceptions Students may incorrectly identify the negative number with the greater absolute value as greater than the negative number with the lower absolute value.	Things to Consider This is the final lesson in the unit. Consider using observations and student reflections in 2.6.5 Wrap-Up to determine what concepts and skills need further work prior to the unit assessment.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk 2.6.1 Warm-Up asks students to identify irrational numbers and order real numbers. Engaging students in a number talk will help them develop their knowledge of and skills with ordering, comparing, and plotting rational and irrational numbers. Ask a variety of students to share their thinking aloud and prompt them to use precise mathematical language to support their explanations.	MA.K12.MTR.3.1: Mathematical Fluency In 2.6.2, 2.6.3, and 2.6.4 Activities, students identify, compare, order, and plot irrational and rational numbers. Repeated practice will help students develop flexibility and accuracy. Use the prompts throughout the activities as evidence of student fluency.
 Differentiate Instruction	Consider having students create a Frayer Model for irrational numbers. This will provide a visual entry point as students see examples, nonexamples, the definition, and visual representations of irrational numbers. This will help students identify irrational numbers in 2.6.3 Activity. Consider reading the prompts aloud in 2.6.5 Wrap-Up to provide an auditory entry point and enable students to focus on reflecting on their mastery of the unit learning targets.	
	Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

Frayer Model

Definition in your own words	Facts/characteristics
Examples	Nonexamples



2.6.1 Warm-Up: Bell Work

Component Overview: Students identify irrational numbers in a set of real numbers. Students order and plot the real numbers on a number line.



2.6.1 Warm-Up: Bell Work

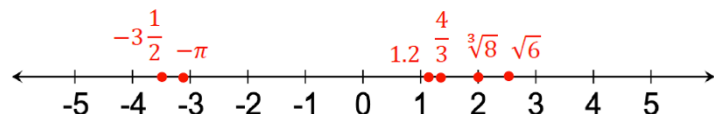
1. Six real numbers are shown.

$$\sqrt{6}, 1.2, -3\frac{1}{2}, \frac{4}{3}, -\pi, \sqrt[3]{8}$$

- Circle all numbers that are irrational.
- Order all the values from greatest to least.

$$\sqrt{6}; \sqrt[3]{8}; \frac{4}{3}; 1.2; -\pi; -3\frac{1}{2}$$

- Plot and label each number on the number line.



Number Talk

Encourage a variety of students to share the strategies they used to identify the irrational numbers and how they ordered and plotted the real numbers. Prompt students to use precise mathematical language as they share their strategies.

2.6.2 Activity: Station 1 – Comparing and Ordering

Component Overview: Students compare and order rational and irrational numbers.



2.6.2 Activity: Station 1 – Comparing and Ordering

1. Compare each pair of expressions using the symbols $<$, $>$, or $=$.

a. -5π ☐ $<$ ☐ $>$ ☐ $=$ $-\sqrt{225}$

b. $\sqrt[3]{125}$ ☐ $<$ ☐ $>$ ☐ $=$ $\frac{\sqrt{100}}{2}$

c. $\sqrt{45}$ ☐ $<$ ☐ $>$ ☐ $=$ $\pi + 4$

d. -7 ☐ $<$ ☐ $>$ ☐ $=$ $-\sqrt[3]{49}$

e. $\frac{29}{3}$ ☐ $<$ ☐ $>$ ☐ $=$ $\sqrt{80}$

f. $\frac{\pi}{3}$ ☐ $<$ ☐ $>$ ☐ $=$ $\sqrt{22} - 3$

2. Choose one pair of expressions from the previous question that were not equivalent.

- a. Circle the pair of expressions.

Answers will vary. Any pair can be circled except for part b. One is circled as an example.

- b. Explain how you determined which value was greater.

Answers will vary. The $\sqrt[3]{49}$ is approximately -3.7 . -3.7 is greater than -7 .

3. Order each set of numbers from least to greatest.

a. $\sqrt{34}$, $\pi + 2$, $5\frac{1}{2}$
 $\pi + 2$; $5\frac{1}{2}$; $\sqrt{34}$

b. $-\sqrt[3]{60}$, $-\frac{18}{4}$, $-\pi$
 $-\frac{18}{4}$; $-\sqrt[3]{60}$; $-\pi$



Students may incorrectly compare the expressions with two negative values. Consider prompting students to consider how these values would be plotted on a number line.

- How would you plot ____ and ____ on a number line?
- What happens to the value of the numbers as you move to the left of zero on a number line?



Consider asking the following scaffolding questions, as needed, of groups while they are working.

- What number is halfway between 25 and 36? How would knowing this help you to compare $5\frac{1}{2}$ and $\sqrt{34}$?
- What number is halfway between 27 and 64? How would knowing this help you to compare $-\pi$ and $-\sqrt[3]{60}$?

2.6.3 Activity: Station 2 – Plotting Rational and Irrational Numbers on a Number Line

Component Overview: Students identify rational and irrational numbers and plot the numbers on a number line.



2.6.3 Activity: Station 2 – Plotting Rational and Irrational Numbers on a Number Line

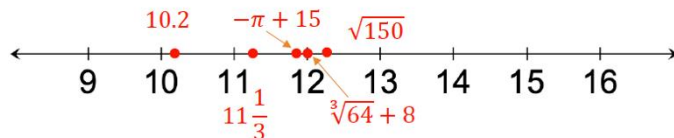
1. Five expressions are shown.

$$\sqrt{150} \quad -\pi + 15 \quad 10.2 \quad 11\frac{1}{3} \quad \sqrt[3]{64} + 8$$

- a. Write each expression under the appropriate description in the table.

Includes Only Rational Numbers	Includes an Irrational Number
10.2 $11\frac{1}{3}$ $\sqrt[3]{64} + 8$	$\sqrt{150}$ $-\pi + 15$

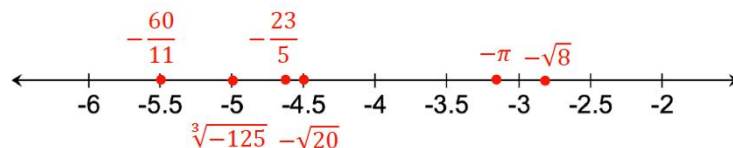
- b. Plot and label each number on the number line.



2. Six expressions are shown.

$$-\sqrt{20} \quad -\frac{60}{11} \quad -\sqrt{8} \quad -\pi \quad \sqrt[3]{-125} \quad -\frac{23}{5}$$

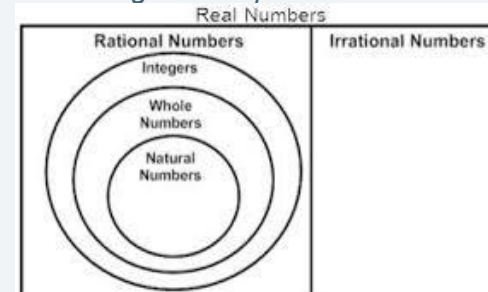
Plot and label the location of each expression on the number line.



Consider creating a larger number line to help students plot the numbers that are in close proximity to each other. If needed, prompt students to partition the number line by tenths using evenly spaced tick marks. The tick marks will help students more accurately plot the numbers.



Consider having students classify the expressions in a Venn diagram template of the real-number system.



Challenge students to create additional expressions to add as examples of each classification of numbers.

2.6.4 Activity: Station 3 – Error Analysis

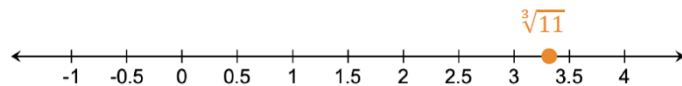
Component Overview: Students approximate nonperfect roots and compare rational and irrational values through error analysis. They also identify and correct errors comparing and plotting real numbers.



2.6.4 Activity: Station 3 – Error Analysis

1. Sara approximated the value of $\sqrt[3]{11}$. Her thinking is shown.

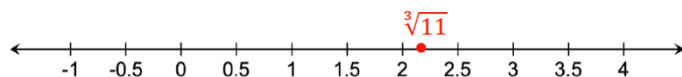
The number 11 is between 9 and 16, which are both perfect squares. So, I know $\sqrt[3]{11}$ lies between 3 and 4 on a number line, but it's closer to 3 because 11 is closer to 9 than 16. I would plot its approximate location here.



- a. Discuss Sara's thinking with your partner. Then, complete the table.

What is correct in Sara's thinking?	What error(s) did Sara make?
Answers will vary. Sara is correct to try to approximate the value of $\sqrt[3]{11}$. Sara is correct that 9 and 16 are perfect squares.	Answers will vary. Sara used perfect squares to approximate the answer instead of perfect cubes. Sara plotted the value in the wrong location.

- b. Plot and label the correct approximate value of $\sqrt[3]{11}$ on the number line.



If students are in small groups as they move through the stations, consider assigning partners within the group so students can have one-on-one discussions about their work. Identifying and correcting errors in samples of student's work will help students avoid making the errors themselves in the future.

2.6.4 Activity: Station 3 – Error Analysis (continued)

2. For each pair of numbers, an explanation comparing the values is provided.
- a. Determine whether each explanation is correct or incorrect.

Pair of Numbers	Comparison and Explanation	Is it Correct?
π^2 and 9	9 is greater than π^2 because π is slightly more than 3, so π^2 is slightly more than 6.	☐ correct ☒ incorrect
$\sqrt{50}$ and 8	$\sqrt{50}$ is less than 8 because 8 is equal to $\sqrt{64}$ and $\sqrt{50}$ is less than that.	☒ correct ☐ incorrect
$\frac{13}{3}$ and $\sqrt[3]{60}$	$\frac{13}{3}$ is greater than $\sqrt[3]{60}$ because $\frac{13}{3}$ is slightly more than 4 and $\sqrt[3]{60}$ is slightly less than 4.	☒ correct ☐ incorrect
-2π and -6	-6 is less than -2π because $-2 \cdot \pi$ is between -6 and -7 , which means it's larger than -6 .	☐ correct ☒ incorrect



Consider asking the following probing questions as groups work.

'Why is _____ incorrect? What is the error? What suggestions would you provide someone to help them avoid the errors?

- b. Two of the explanations above are incorrect. Write the correct explanation for each of them by completing the sentence stems.

-6 is greater than -2π because *$-2 \cdot \pi$ is between -6 and -7 , which means it's smaller than -6 .*

9 is less than π^2 because ... *because π is slightly more than 3, π^2 is slightly more than 9.*

Note: these could be given in the reverse order.

2.6.5 Wrap-Up: Reflection

Component Overview: Students reflect on their learning throughout the unit.

2.6.5 Wrap-Up: Reflection

- Circle the icon that best describes your understanding of each concept learned in this unit.

Answers will vary.



I need help.



I got it.



I could help someone.

Solve equations in the form $x^2 = p$.	
Solve equations in the form $x^3 = q$.	
Calculate square roots of perfect squares and cube roots of perfect cubes.	
Identify whether a number is rational or irrational.	
Locate the approximate value of expressions involving irrational numbers on a number line.	
Plot, order, and compare rational and irrational numbers on a number line.	



Consider reading the prompts aloud so students can focus on reflecting on their level of mastery with each skill. Consider sharing concrete examples for each statement. For example, instead of asking if students can calculate square roots of perfect squares and cube roots of perfect cubes, ask students if they can calculate $\sqrt{16}$ and $\sqrt[3]{27}$.